

KCPC-CS111

Supplementary Reading

Week 1: Introduction to Competitive Programming, Invariants, and Logic

King's Competitive Programming Club

This document contains curated supplementary readings for Workshop 1. These texts expand upon the core themes introduced in the session: competitive programming fundamentals, invariant reasoning, epistemic logic (common knowledge), and impartial combinatorial games.

I. Competitive Programming Fundamentals

Competitive Programmer's Handbook (Antti Laaksonen)

<https://cses.fi/book/book.pdf>

- **Chapter 1: Introduction** Contest environment, basic language features, and input/output optimisation.
- **Chapter 2: Time Complexity** Asymptotic notation (O -notation), estimating operation counts per second (10^8 ops/sec rule), and detecting TLE bounds.

Competitive Programming 4 (Steven Halim, Felix Halim, Suhendry)

<https://cpbook.net/>

- **Chapter 1: Introduction** Contest categories (ICPC, IOI, Codeforces, AtCoder), standard submission verdicts, and typical problem pitfalls.
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II. Invariants and Mathematical Heuristics

Problem-Solving Strategies (Arthur Engel)

<https://link.springer.com/book/10.1007/978-0-387-21607-2>

- **Chapter 1: The Invariance Principle** Foundational treatment of invariants, monovariants, and coloring arguments in discrete dynamical processes.

The Art and Craft of Problem Solving (Paul Zeitz)

John Wiley & Sons, 2006.

- **Chapter 3: Tactics for Solving Problems** Symmetry, parity, extremal thinking, and simplifying assumptions.
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III. Epistemic Logic and Common Knowledge

Reasoning About Knowledge (Fagin, Halpern, Moses, Vardi)

MIT Press, 2003. <https://mitpress.mit.edu/9780262562003/reasoning-about-knowledge/>

- **Chapter 1: Introduction** The distinction between mutual knowledge and common knowledge; the formal mud-on-the-forehead puzzle.
 - **Chapter 2: A Model for Knowledge** Kripke structures and epistemic possible worlds.
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IV. Combinatorial Game Theory

Game Theory (Thomas S. Ferguson)

https://www.math.ucla.edu/~tom/Game_Theory/comb.pdf

- **Part I: Impartial Combinatorial Games** Subtraction games, normal-play conventions, N - and P -positions, and backward induction on directed acyclic graphs.

CP-Algorithms: Games on Arbitrary Graphs

https://cp-algorithms.com/game_theory/games_on_graphs.html

- Topological determination of winning and losing states via retro-analysis / backward induction.